

IPL DATA ANALYSIS PROJECT

Python + Power BI Performance Analytics

Project Report

Analysis period: 2008-2017

Matches	636
Total Runs	194,313
Teams	13
Players	495

Prepared as a portfolio-style data analytics project report

1. Project Overview

Objective. Analyze IPL match and ball-by-ball data to understand team performance, player performance, season trends, toss outcomes, venue usage, scoring patterns, and bowling performance, and present the findings in an interactive Power BI dashboard.

- **Business focus:** identify high-performing teams and players, season-level changes, and match-level patterns.
- **Analytical approach:** load and clean the source CSV files in Python, perform exploratory analysis and visualizations, then build an interactive Power BI model and dashboard.
- **Final outputs:** cleaned CSV files, Python analysis script, Power BI dashboard, report, and project documentation.

2. Objectives

- Measure overall IPL activity and scoring across the available seasons.
- Compare team wins, average batting output, and venue usage.
- Identify leading run scorers, wicket takers, and Player of the Match award leaders.
- Study toss decisions and the relationship between winning the toss and winning the match.
- Analyze season-wise runs and wickets, scoring distributions, and match-level runs versus wickets.
- Create an interactive dashboard with Season, Team, and Venue slicers.

3. Dataset Description

File	Original rows	Original columns	Purpose
matches.csv	636	18	One row per match: teams, toss, winner, venue, player of match, margins, date.
deliveries.csv	150,460	21	Ball-by-ball data: batting team, bowling team, batsman, bowler, runs, extras, dismissals.

Final cleaned tables used in Power BI: IPL_Matches_Cleaned.csv (636 rows, 17 columns) and IPL_Deliveries_Cleaned.csv (150,459 rows, 22 columns, including Season).

4. Data Preparation and Cleaning

- Inspected shape, columns, data types, descriptive statistics, missing values, duplicates, and categorical values.
- Removed the one exact duplicate delivery row.
- Standardized the inconsistent team label “Rising Pune Supergiants” to “Rising Pune Supergiant”.
- Converted the match date column to a date/datetime type.
- Removed the empty umpire3 column from the matches table.
- Retained structural missing values in delivery-level dismissal and fielder fields rather than filling them with artificial values.
- Merged Season into the deliveries table for season-based delivery analysis and removed the merge-added duplicate match id before final export.
- Rechecked duplicates and final shapes before exporting the cleaned CSV files.

Data quality note. Three matches do not have a recorded winner. For toss-win-rate analysis, the denominator is therefore 633 matches with a recorded winner.

5. Python Analysis

Tools used: Python, Pandas, NumPy, Matplotlib, and Seaborn.

Analysis area	Main calculation / question
Overall activity	Total matches and matches per season.
Team performance	Wins by team, highest/lowest win counts, team-wise average runs.
Batting	Top run scorers, top individual scores, season-wise total runs.
Bowling	Bowler-credited wickets and top wicket takers.
Toss	Toss decisions and whether the toss winner also won the match.
Venue	Most frequently used venue and city.
Match outcomes	Largest winning margins by runs and wickets.
Relationships	Runs vs wickets scatter and correlation analysis.

Bowler-credited wicket definition. The Power BI wicket KPI and bowler ranking use caught, bowled, lbw, stumped, caught and bowled, and hit wicket. This excludes dismissals that are not credited to the bowler, such as run-outs.

6. Power BI Data Model and Dashboard

Core relationship: IPL_Matches_Cleaned[id] (one) -> IPL_Deliveries_Cleaned[match_id] (many).

Team filtering model: Teams -> MatchTeams -> IPL_Matches_Cleaned -> IPL_Deliveries_Cleaned.
This lets one Team slicer filter matches and delivery-level visuals.

Measure	Definition / result
Total Matches	COUNTROWS of the cleaned matches table; 636.
Total Runs	SUM of delivery total_runs; 194,313.
Total Wickets	Bowler-credited wicket rows; 6,673.
Total Teams	Distinct teams from team1/team2; 13.
Total Players	Distinct names appearing as batsman or bowler; 495.
Average Runs	Total Runs / Total Matches; 305.52 runs per match.
Average Winning Margin	Dashboard measure combining run-margin and wicket-margin values; 17.14. Interpret with caution because runs and wickets are different units.
Toss Winner Win %	Toss winner also won 325 of 633 matches with a recorded winner; 51.34%.

Dashboard pages: IPL Performance Analytics Dashboard (overview), IPL Detailed Analysis (detailed visuals and slicers), and IPL Report (key insights).

7. Dashboard Snapshot - Overview

The overview page combines the main KPI cards with season, team, batting, and bowling comparisons.

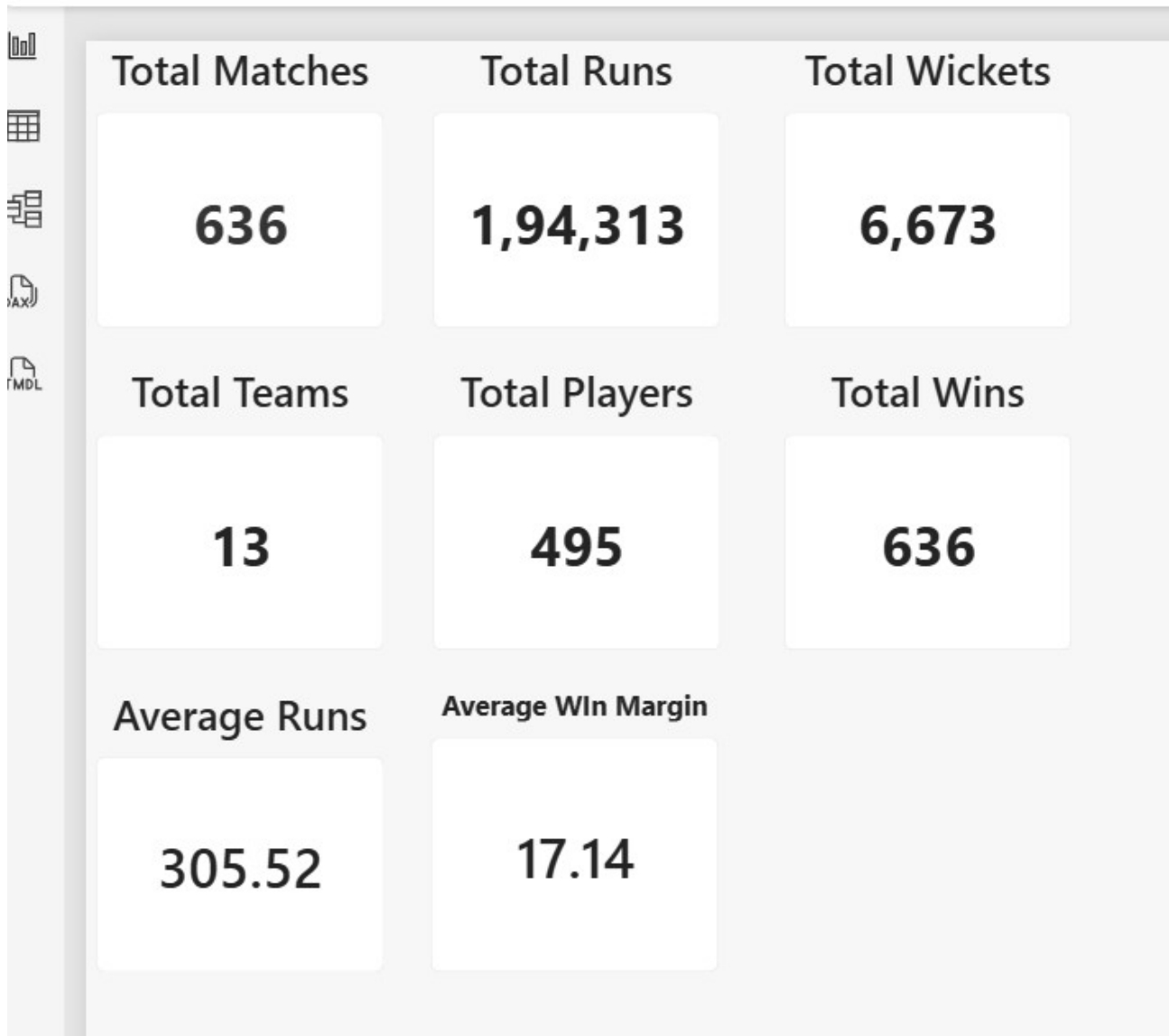


Figure 1. IPL overview dashboard showing the eight KPI cards.

8. Detailed Analysis Dashboard

The detailed page contains the remaining analytical visuals and supports interactive filtering by Season, Team, and Venue.

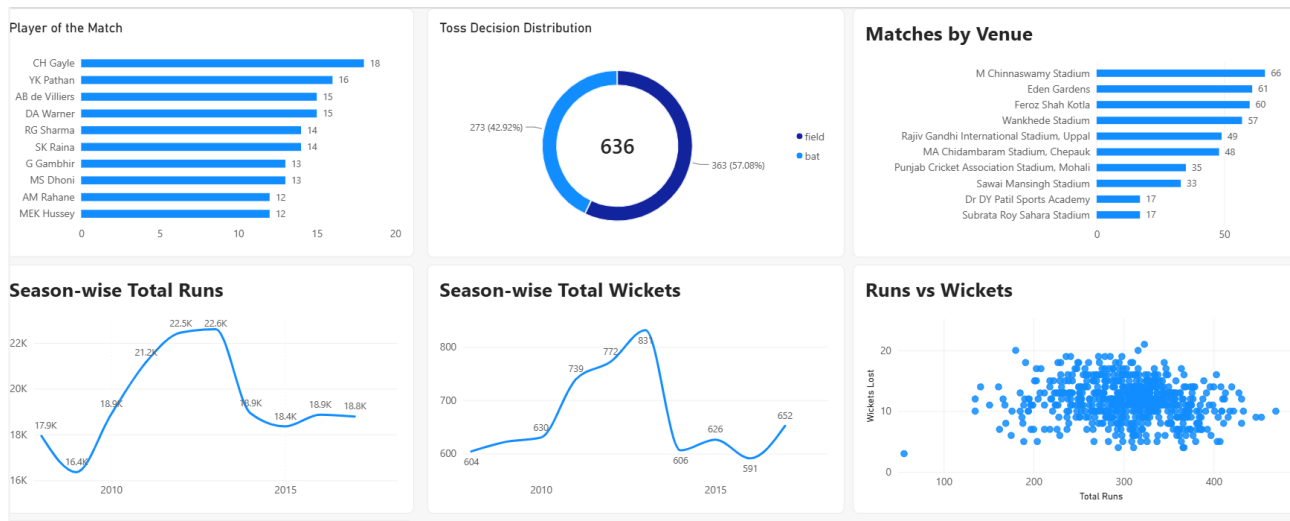


Figure 2. IPL detailed analysis visuals, including season trends, venue usage, scatter analysis, and correlation heatmap.

9. Key Insights and Findings

- **Overall dataset:** 636 matches were analyzed across 2008-2017, with 194,313 total runs, 6,673 bowler-credited wickets, 13 teams, and 495 unique player names.
- **Season performance:** 2013 had the highest number of matches (76), the highest total runs (22,602), and the highest bowler-credited wicket count (831).
- **Team performance:** Mumbai Indians recorded the highest team win count in the cleaned analysis (92). Chennai Super Kings had the highest average runs per batting innings at approximately 158.33, followed by Gujarat Lions (156.84) and Mumbai Indians (155.19).
- **Batting performance:** SK Raina led the run-scoring table with 4,548 runs, followed by V Kohli (4,423) and RG Sharma (4,207). The highest individual score in the dataset was 175 by CH Gayle.
- **Bowling performance:** SL Malinga led bowler-credited wickets with 154, followed by A Mishra (134) and Harbhajan Singh (127).
- **Player of the Match:** CH Gayle received the highest number of Player of the Match awards in the analysis, with 18.
- **Toss analysis:** Teams chose to field first 363 times and bat first 273 times. The toss winner also won 325 of 633 matches with a recorded winner, a 51.34% win rate.
- **Venue analysis:** M Chinnaswamy Stadium was the most frequently used venue, hosting 66 matches in the analyzed data.
- **Scoring:** Average scoring across matches was approximately 305.52 total runs per match.

10. Conclusion

The project demonstrates a complete data analytics workflow: raw CSV ingestion, quality checks, cleaning and standardization in Python, exploratory calculations and visual analysis, relational modeling in Power BI, DAX-based KPI creation, and interactive dashboard delivery. The combined

Python and Power BI workflow makes it possible to move from raw ball-by-ball data to business-friendly performance insights.

- Python was used for reproducible data cleaning, calculations, and analytical visualizations.
- Power BI was used for interactive reporting, filtering, KPI cards, and dashboard storytelling.
- The final project emphasizes repeatable analysis and clear communication of findings rather than only producing charts.

11. Final Deliverables

Folder / file	Purpose
Dataset/IPL_Matches_Cleaned.csv	Cleaned match-level dataset for Power BI and analysis.
Dataset/IPL_Deliveries_Cleaned.csv	Cleaned ball-by-ball dataset for Power BI and analysis.
Python/IPL_Data_Analysis.py	Python workflow for cleaning, analysis, visualizations, and export.
PowerBI/IPL_Analytics_Dashboard.pbix	Interactive Power BI dashboard and report pages.
Report/IPL_Project_Report.pdf	Project documentation and findings.
README.md	Project overview, workflow, tools, and usage notes.

12. Notes for Portfolio Presentation

- Lead with the business questions and decisions the dashboard answers.
- Explain why Python was used for cleaning and reproducibility, while Power BI was used for interactive reporting.
- When discussing wickets, specify that the KPI uses bowler-credited dismissals.
- When discussing toss impact, emphasize that the 51.34% result indicates only a modest advantage over an even 50% in this dataset.
- Keep the final submission folder organized so recruiters or reviewers can open the PBIX, inspect the Python script, and read the PDF without searching for files.

End of Report